C8 — Risk controls, denoising and new quoting ideas, on a month split

Not a ladder phase. It comes from the Part C follow-up list after C4, with the forward test left out on request. Lab notebook: `experiments/c8_risk_controls/README.md`.

> **At a glance**
>
> - **The best result in Part C is OBI on SUIUSDC:** +2.52% over seven held-out sessions at 175 fills a day, positive on 6 of 7. It stays positive **without the 09-22 crash** (+1.77%), **without its best session** (+1.52%), **without the maker rebate** (+2.05%), **with a 1 bp maker fee** (+1.13%), at **4× order latency** (+1.30%) and under the **pessimistic queue model** (+2.46%); no other configuration passes all six. Its one weak point is halving the markout κ (−0.90%).
> - **OBI on XRPUSDC survives every cost and latency stress** (+1.68% with a 2 bp maker fee, +1.17% at 4× latency), **but its held-out gain is the crash session**: +0.09% without it. OBI on ETHUSDC's +0.85% is the maker rebate: +0.09% without it, −2.92% at a 2 bp fee.
> - **The circuit breaker is the one control that pays on ETH.** It raises held-out return on all three ETH strategies (+0.9 to +2.8 points), and with it GLFT goes from −3.25% to −0.72%. On OBI XRP it costs 2.3 points.
> - **Adding more knobs hurts the strategies that already work.** A greedy combination chosen on training sessions improves the losing strategies and makes OBI worse on both assets (ETH +0.85 → +0.48, XRP +2.84 → +1.49).
> - **RL on top of the markout rule never beats the rule**: worse on all three ETH strategies by 0.5–1.5 points, a tie on OBI XRP (+2.83 against +2.84), with seed ranges of 1.3 to 9.7 points.

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1. The task

The request, after C4:

> do all part C follow-ups, except 4 [the forward test]; use as much training days as possible and less held-out days (they together should cover month period); do we have denoising of signals already; make sure all key tables and plots presented in writeups; do not reject or estimate any ideas without real stress test because of nature of part C; do you have any new ideas how we can improve current strategies?

The follow-ups were: a **circuit breaker**, a **daily loss limit**, **stress tests** of the one configuration that made money, the **breadth assets** on more held-out sessions, **RL on top of the markout rule**, and the **residual-LayerNorm network** on a GPU (reported in C4 §8). To those C8 adds **denoising** of the signals and eight **new quoting ideas**, each run through the backtest rather than argued about, and a **stress test** of every configuration reported as a result.

2. The month split and how it was run

The data covers 2025-08-26 to 2025-09-25. C8 trains on **24 sessions** (08-26 to 09-18) and holds out **7** (09-19 to 09-25), a full week that keeps a weekend and the 09-22 crash. Earlier phases held out eleven sessions; their numbers are recomputed here on the same last seven wherever they are compared.

Every configuration is replayed once per session and cached by a hash of the configuration, so the split, the walk-forward selection and the stress tests all read the same results (`experiments/c8_risk_controls/c8.py`). The stages, each on top of the previous stage's choice for each (strategy, asset):

stage	what	chosen on
A	floor × markout κ × skew, C4's grid with κ = 1 added: 24 points	training sessions
B	circuit breakers and loss limits: 25 variants	training sessions
C	denoising and new ideas: up to 27 variants	training sessions
D	greedy combination of each family's best variant	training sessions
E	stress tests of the A and D choices	(reported only)

A choice must take at least 20 fills on a median training session, so a configuration cannot win by not trading. In all, about 36,000 session replays.

3. The base grid on 24 training sessions

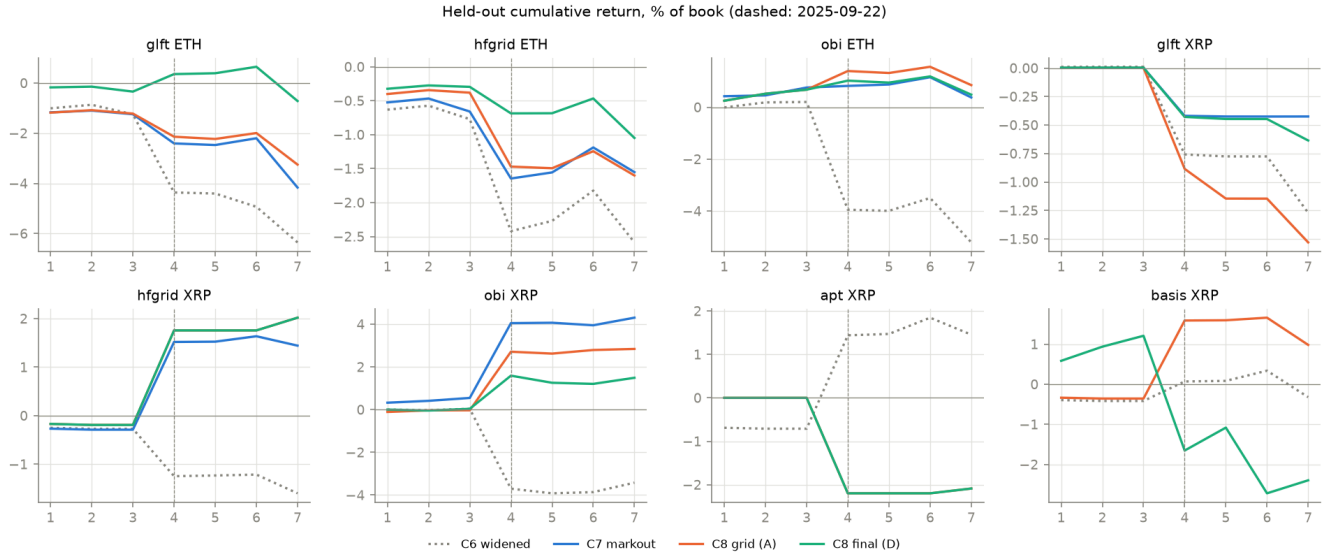


Figure — held-out cumulative return over the seven sessions. Dotted: C6's widened preset; blue: C7's markout rule (κ chosen on four sessions); orange: the C8 grid choice; green: the C8 final combination (§5). Dashed line: 2025-09-22.

asset	strategy	C6 widened	C7 markout	C8 grid	training	without 09-22	sessions up	fills/day
ETH	glft	−6.35	−4.17	**−3.25**	+3.37	−2.34	2/7	27
ETH	hfgrid	−2.59	−1.55	**−1.60**	+1.93	−0.51	2/7	22
ETH	obi	−5.21	+0.38	**+0.85**	+3.37	+0.14	5/7	369
XRP	glft	−1.27	−0.42	**−1.53**	−2.94	−0.65	2/7	4
XRP	hfgrid	−1.60	+1.44	**+2.02**	+1.78	+0.07	2/7	1
XRP	obi	−3.42	+4.31	**+2.84**	+1.41	+0.09	5/7	106
XRP	apt	+1.45	–	**−2.09**	+3.01	+0.11	1/7	0
XRP	basis	−0.32	–	**+0.98**	+3.20	−0.95	3/7	4

Held-out cumulative return, % of book. "training" is the chosen point's return over the 24 training sessions.

The chosen points:

asset	strategy	floor	κ	skew
ETH	glft	4 bp (×1)	2	×1
ETH	hfgrid	4 bp (×1)	1	×1
ETH	obi	6 bp (×1.5)	4	×1
XRP	glft	8 bp (×1)	2	×2
XRP	hfgrid	12 bp (×1.5)	1	×1
XRP	obi	12 bp (×1.5)	2	×2
XRP	apt	16 bp (×2)	0	×1

asset	strategy	floor	κ	skew
XRP	basis	8 bp (×1)	2	×1

More training data changed the XRP OBI choice from C7's (floor 8 bp, κ = 1) to a wider one, and on this held-out week the wider choice earns less: +2.84 against +4.31. On ETH the 24-session choices beat C7's on `glft` and `obi`. Four of the eight choices (all on XRP) trade four times a day or less; they are reported, but not as market making.

4. Risk controls, denoising and new ideas, one family at a time

4.1 The families

All are in C6's quoting kernel (`zoo.py`), off by default; an old markout run replays to the identical return and fill count with them compiled in.

family	the idea	variants
circuit breaker	pause quoting when 10 s volatility exceeds 2.5× or 4× its 30 min level, or when aggressive flow is one-sided (imbalance above 0.5 or 0.7) while above its normal level; both sides or only the side being hit; 30 s or 5 min	16
loss limit	reduce-only after the session loses 25, 50 or 100 bp; or after a 15, 30 or 60 bp drawdown from the session's peak, for 10 min or 1 h	9
signal smoothing	exponential average of the fair-price signal over 1 s, 5 s or 20 s	3
alpha clipping	clip OBI's standardised imbalance at ±1.5 or ±2.5	2
markout half-life	the toxicity average over 5 or 50 fills instead of 20	2
momentum	lean the fair price along the last 1 s or 5 s of mid drift, ×0.5, ×1 or ×2	6
per-side markout	one toxicity average per side; widen only the side whose fills were toxic	1
requote hysteresis	keep a resting quote while the new one is within 1 or 2 grid steps, to keep queue position	2
toxicity sizing	shrink order size as toxicity rises instead of only widening	2
inventory flatten	above 50% or 80% of the position cap, cut back to half with an IOC at the touch	2
time of day	×2 floor in one four-hour UTC block; used as a schedule of every block that helped	6

4.2 What each family adds

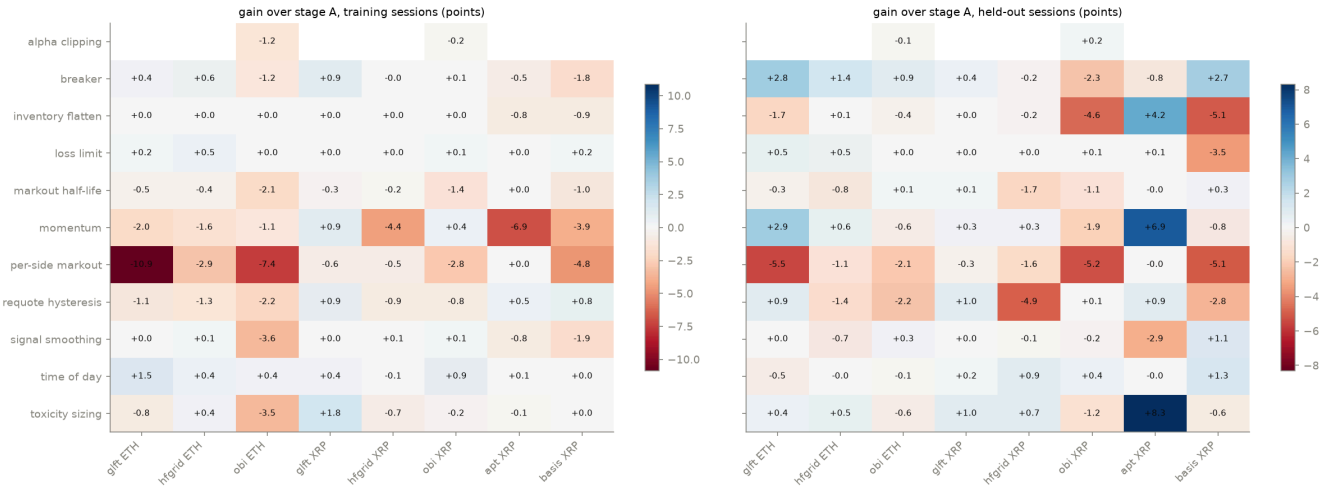


Figure — each family's best variant (chosen on training sessions) against the stage-A choice: gain on training sessions (left) and held out (right), in points of return.

family	ETH glft	ETH hfgrid	ETH obi	XRP glft	XRP hfgrid	XRP obi	XRP apt	XRP basis
circuit breaker	**+0.39 / +2.80**	**+0.60 / +1.41**	−1.18 / +0.90	+0.95 / +0.43	−0.03 / −0.24	+0.05 / −2.28	−0.53 / −0.84	−1.81 / +2.72
loss limit	+0.16 / +0.54	+0.49 / +0.50	+0.00 / +0.00	+0.00 / +0.00	+0.00 / +0.00	+0.11 / +0.06	+0.00 / +0.05	+0.19 / −3.45
time of day	+1.48 / −0.45	+0.44 / −0.02	+0.38 / −0.11	+0.37 / +0.16	−0.05 / +0.85	+0.92 / +0.36	+0.08 / −0.00	+0.01 / +1.33
toxicity sizing	−0.83 / +0.40	+0.39 / +0.54	−3.55 / −0.55	+1.82 / +0.96	−0.72 / +0.73	−0.18 / −1.17	−0.13 / +8.31	+0.04 / −0.62
requote hysteresis	−1.10 / +0.92	−1.26 / −1.38	−2.25 / −2.22	+0.89 / +0.95	−0.85 / −4.94	−0.78 / +0.06	+0.47 / +0.88	+0.80 / −2.77
momentum	−2.04 / +2.90	−1.59 / +0.64	−1.14 / −0.57	+0.94 / +0.32	−4.44 / +0.31	+0.44 / −1.93	−6.95 / +6.93	−3.90 / −0.78
signal smoothing	+0.00 / +0.00	+0.09 / −0.72	−3.56 / +0.26	+0.00 / +0.00	+0.08 / −0.07	+0.09 / −0.22	−0.84 / −2.95	−1.87 / +1.14
alpha clipping	–	–	−1.24 / −0.08	–	–	−0.16 / +0.17	–	–
markout half-life	−0.47 / −0.31	−0.40 / −0.78	−2.11 / +0.15	−0.31 / +0.08	−0.16 / −1.67	−1.41 / −1.06	+0.00 / −0.00	−1.04 / +0.26
inventory flatten	+0.00 / −1.67	+0.00 / +0.05	+0.03 / −0.37	+0.00 / +0.00	+0.00 / −0.15	+0.00 / −4.64	−0.80 / +4.19	−0.90 / −5.06
per-side markout	−10.87 / −5.46	−2.94 / −1.10	−7.44 / −2.15	−0.63 / −0.27	−0.54 / −1.60	−2.81 / −5.24	+0.00 / −0.00	−4.80 / −5.07

Training / held out, points of return over the stage-A choice. XRP hfgrid, apt and basis trade 0–4 times a day, so their cells are noise.

- **The circuit breaker is the only family that raises held-out return on all three ETH strategies**, and on `glft` and `hfgrid` it does so on training sessions too, so it can be chosen honestly. On OBI XRP it costs 2.3 points: there the crash session is where the money is made, and a breaker pauses exactly then.
- **The loss limit helps a little and consistently where it triggers** (+0.5 points on ETH `glft` and `hfgrid`), and barely moves the OBI choices (0 on ETH, +0.06 on XRP).
- **Momentum pays held out on ETH `glft` (+2.90) but loses on training sessions (−2.04)**, so no honest selection picks it; time of day is the reverse, a training-session gain that does not hold.
- **Per-side markout is the clearest failure**: worse everywhere it can be measured, by up to 11 points on training sessions. Widening only the side that was hit leaves the other side quoting into the same move.

4.3 Denoising, specifically

The question was whether the signals are denoised already. Before C8, only in two places: C7's markout rule is itself an exponential average over the last 20 fills, and OBI's imbalance is standardised over a ten-minute window but not smoothed. C8 tested three denoisers:

denoiser	best held-out gain on a strategy that trades	where it hurt
signal smoothing (1–20 s)	+0.26 on ETH OBI, chosen at −3.56 on training	−0.72 ETH hfgrid
alpha clipping (±1.5, ±2.5 σ)	+0.17 on XRP OBI	−1.24 on ETH OBI training
markout average over 5 or 50 fills	+0.15 on ETH OBI	−1.06 XRP OBI, −1.67 XRP hfgrid

None of the three improves any strategy on both training and held-out sessions, so the combination step (§5) never picks one for a strategy that trades. The denoising that is in use is inside the circuit breaker: its triggers compare fast and slow exponential averages, require above-normal activity, and hold for a cooldown, so a single print cannot trip it.

5. Combining them

Stage D adds, one family at a time, whichever family's best variant most improves the training-session return of the current choice, and stops when none does.

asset	strategy	C8 grid	final	without 09-22	fills/day	added (in order)
ETH	glft	−3.25	**−0.72**	−1.41	25	circuit breaker (volatility 2.5×, both sides, 30 s), markout over 50 fills, ×2 floor 08:00–16:00 UTC
ETH	hfgrid	−1.60	**−1.05**	−0.66	16	×2 floor 16:00–24:00 UTC, daily loss limit 25 bp
ETH	obi	+0.85	**+0.48**	+0.14	369	inventory flatten above 80% of the cap
XRP	glft	−1.53	**−0.64**	−0.20	36	toxicity sizing, per-side markout

asset	strategy	C8 grid	final	without 09-22	fills/day	added (in order)
XRP	hfgrid	+2.02	+2.02	+0.07	1	nothing improved training
XRP	obi	+2.84	**+1.49**	−0.06	95	momentum, toxicity sizing, trailing loss limit
XRP	apt	−2.09	−2.09	+0.11	0	nothing improved training
XRP	basis	+0.98	**−2.40**	+0.44	317	requote hysteresis, signal smoothing

Held-out cumulative return, %. The final choices are also the green lines in the §3 figure.

The combination helps the strategies that lose — ETH `glft` gains 2.5 points and XRP `glft` 0.9 — and hurts the ones that make money: OBI loses 0.4 points on ETH and 1.4 on XRP. Every addition is chosen on 24 training sessions and each one fits them; with the strongest strategy, the extra parameters fit the training month rather than the market.

6. Stress tests

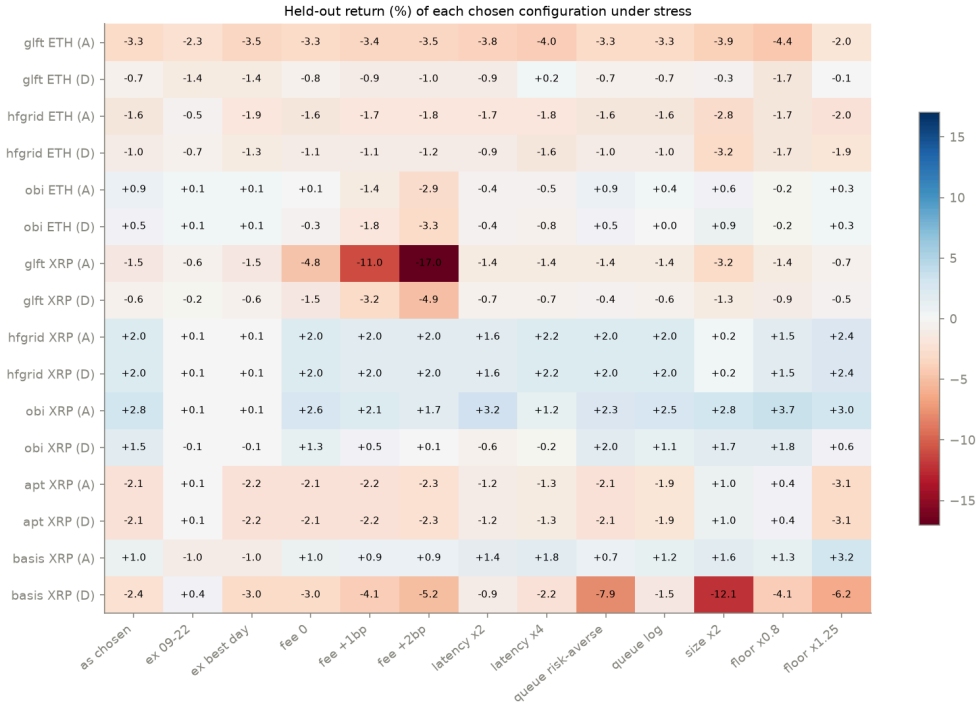


Figure — held-out return of each chosen configuration, as chosen and under each stress (A: the grid choice; D: the final combination).

asset	strategy	choice	as chosen	fills/day	ex 09-22	ex best day	fee 0	fee +1bp	fee +2bp	latency ×2	latency ×4	queue risk-averse	queue log	size ×2	floor ×0.8	floor ×1.25	κ ×0.5	κ ×2
ETH	glft	A	−3.25	27	−2.34	−3.49	−3.29	−3.38	−3.46	−3.78	−4.01	−3.25	−3.25	−3.88	−4.40	−1.97	−3.96	−1.86
ETH	glft	D	−0.72	25	−1.41	−1.41	−0.77	−0.86	−0.96	−0.92	+0.20	−0.72	−0.72	−0.30	−1.67	−0.08	+1.12	−0.67
ETH	hfgrid	A	−1.60	22	−0.51	−1.85	−1.65	−1.73	−1.82	−1.74	−1.81	−1.60	−1.60	−2.80	−1.71	−2.00	−2.10	−1.83
ETH	hfgrid	D	−1.05	16	−0.66	−1.26	−1.07	−1.13	−1.18	−0.90	−1.65	−1.05	−1.05	−3.24	−1.66	−1.91	−2.16	−1.78
ETH	obi	A	+0.85	369	+0.14	+0.14	+0.09	−1.42	−2.92	−0.40	−0.48	+0.91	+0.37	+0.64	−0.17	+0.26	−0.58	+0.34
ETH	obi	D	+0.48	369	+0.14	+0.14	−0.28	−1.81	−3.31	−0.40	−0.78	+0.55	+0.00	+0.94	−0.17	+0.27	−0.37	+0.01
XRP	glft	A	−1.53	4	−0.65	−1.53	−4.77	−11.04	−17.02	−1.40	−1.41	−1.41	−1.38	−3.18	−1.45	−0.67	−1.54	−0.89
XRP	glft	D	−0.64	36	−0.20	−0.64	−1.51	−3.24	−4.95	−0.73	−0.70	−0.41	−0.61	−1.25	−0.91	−0.48	−0.52	−0.49
XRP	hfgrid	A = D	+2.02	1	+0.07	+0.07	+2.01	+1.99	+1.97	+1.57	+2.21	+2.02	+2.02	+0.24	+1.52	+2.39	+1.39	+2.67
XRP	obi	A	**+2.84**	106	+0.09	+0.09	**+2.61**	**+2.14**	**+1.68**	**+3.22**	**+1.17**	**+2.27**	**+2.49**	+2.83	+3.71	+2.98	+3.57	+2.80

asset	strategy	choice	as chosen	fills/day	ex 09-22	ex best day	fee 0	fee +1bp	fee +2bp	latency ×2	latency ×4	queue risk-averse	queue log	size ×2	floor ×0.8	floor ×1.25	κ ×0.5	κ ×2
XRP	obi	D	+1.49	95	−0.06	−0.06	+1.28	+0.45	+0.13	−0.63	−0.17	+1.98	+1.09	+1.67	+1.76	+0.62	+1.78	+0.91
XRP	apt	A = D	−2.09	0	+0.11	−2.20	−2.13	−2.20	−2.28	−1.16	−1.35	−2.09	−1.88	+1.01	+0.44	−3.06	–	–
XRP	basis	A	+0.98	4	−0.95	−0.95	+0.96	+0.93	+0.89	+1.41	+1.76	+0.69	+1.18	+1.57	+1.32	+3.19	+0.95	+0.38
XRP	basis	D	−2.40	317	+0.44	−2.96	−2.96	−4.07	−5.17	−0.94	−2.16	−7.91	−1.47	−12.14	−4.12	−6.19	−3.88	−2.87

Held-out cumulative return, %. The default maker fee is Binance's −0.5 bp rebate; "fee 0" removes it. Latency multiplies the calibrated order-latency model; "risk-averse" and "log" are hftbacktest's more pessimistic queue models; "size ×2" doubles the order size at the same position cap.

What the stresses separate:

1. **OBI ETH's +0.85% is the rebate.** At 369 fills a day the −0.5 bp rebate is worth about 0.65% over the week; without it the return is +0.09%, at a 1 bp fee −1.42%, and doubled latency or a 20% narrower floor also turn it negative. ETH still has no configuration that makes money robustly.
2. **OBI XRP survives every cost and latency stress** — +1.68% at a 2 bp fee, +1.17% at 4× latency, +2.27% under the risk-averse queue — **and fails the one that matters most**: without 09-22, which is also its best session, it is +0.09%. It makes money by holding the right side through a crash, not by earning the spread on ordinary days.
3. **The circuit breaker's ETH GLFT gain is not a cost artefact**: −0.72% stays between −0.3% and −1.0% under every fee and queue stress. It is still a loss.
4. **XRP GLFT's grid choice depends on the rebate** (−1.53% → −17.02% at a 2 bp fee), which is what a strategy looks like when its few fills are worth less than the rebate paid on them.

7. The breadth assets

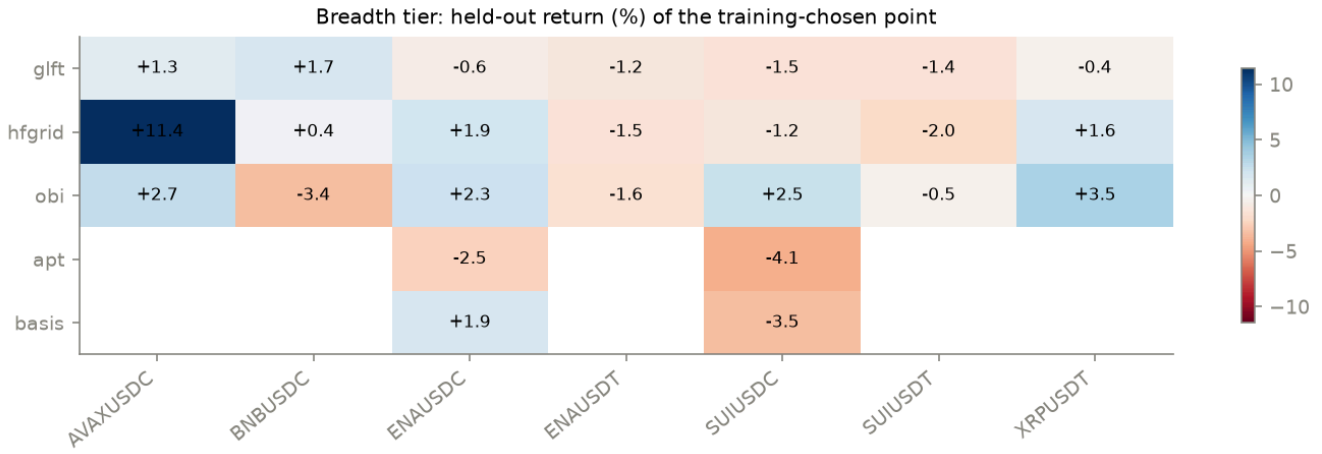


Figure — held-out return of the training-chosen point, per asset and strategy (floor × κ grid, skew at preset).

asset	strategy	chosen point	train	held out	ex-09-22	days up	fills/day
XRPUSDT	glft	floor 12 bp, κ 1	+0.46	−0.42	−0.00	0/7	0
XRPUSDT	hfgrid	floor 12 bp, κ 4	+2.91	+1.61	−0.25	1/7	0

asset	strategy	chosen point	train	held out	ex-09-22	days up	fills/day
XRPUSDT	**obi**	floor 8 bp, κ 2	+4.79	**+3.53**	+0.48	5/7	189
SUIUSDC	glft	floor 8 bp, κ 4	−4.91	−1.50	−0.96	3/7	1168
SUIUSDC	hfgrid	floor 8 bp, κ 1	+5.01	−1.16	−0.59	4/7	11
SUIUSDC	**obi**	floor 8 bp, κ 4	+3.60	**+2.52**	**+1.77**	6/7	175
SUIUSDC	apt	floor 8 bp, κ 2	+12.31	−4.10	−3.09	4/7	44
SUIUSDC	basis	floor 8 bp, κ 0	+4.41	−3.45	−1.48	4/7	16
SUIUSDT	glft	floor 8 bp, κ 4	+0.37	−1.35	−0.83	3/7	24
SUIUSDT	hfgrid	floor 12 bp, κ 4	+7.75	−2.03	−1.13	1/7	4
SUIUSDT	obi	floor 16 bp, κ 4	+3.65	−0.53	+1.90	5/7	44
ENAUSDC	glft	floor 32 bp, κ 4	−7.19	−0.56	−0.53	0/7	154
ENAUSDC	hfgrid	floor 32 bp, κ 4	+1.55	+1.91	+0.25	3/7	2
ENAUSDC	obi	floor 32 bp, κ 4	+3.73	+2.30	+0.68	5/7	2
ENAUSDC	apt	floor 32 bp, κ 2	+28.10	−2.55	+0.65	3/7	16
ENAUSDC	basis	floor 32 bp, κ 0	+4.35	+1.86	+0.37	3/7	3
ENAUSDT	glft	floor 32 bp, κ 4	−1.03	−1.16	−1.28	2/7	791
ENAUSDT	hfgrid	floor 48 bp, κ 1	+1.93	−1.51	+0.02	1/7	0
ENAUSDT	obi	floor 32 bp, κ 2	+12.36	−1.56	−1.13	2/7	6
AVAXUSDC	glft	floor 24 bp, κ 1	+0.25	+1.32	−0.08	2/7	12
AVAXUSDC	hfgrid	floor 16 bp, κ 1	+1.36	+11.44	+3.11	6/7	17
AVAXUSDC	**obi**	floor 16 bp, κ 0	+7.30	**+2.68**	**+2.34**	4/7	150
BNBUSDC	**glft**	floor 2 bp, κ 1	+1.71	**+1.72**	+1.53	6/7	489
BNBUSDC	hfgrid	floor 3 bp, κ 4	+1.76	+0.36	+0.04	4/7	27
BNBUSDC	obi	floor 7.4 bp, κ 4	+3.04	−3.40	−2.67	2/7	524

The seven other assets on the same split: floor × κ per strategy, skew at preset, C7's σ correction (×1.445, measured on ETH and XRP) transferred to `glft` and `hfgrid`.

OBI makes money held out on three of the seven breadth assets (XRPUSDT, SUIUSDC, AVAXUSDC) at 150–190 fills a day — five of all nine with ETHUSDC and XRPUSDC — and on all three it stays positive without the crash. Everything positive and trading at least 20 times a day was stress-tested:

asset	strategy	as chosen	fills/day	ex 09-22	ex best day	fee 0	fee +1bp	fee +2bp	latency ×2	latency ×4	queue risk-averse	queue log	size ×2	floor ×0.8	floor ×1.25	κ ×0.5	κ ×2
SUIUSDC	obi	**+2.52**	175	**+1.77**	**+1.52**	**+2.05**	**+1.13**	+0.21	**+1.86**	**+1.30**	**+2.46**	**+2.17**	+2.74	+1.43	+1.36	−0.90	+0.56
AVAXUSDC	obi	+2.68	150	+2.34	−0.37	+2.35	+1.69	+1.04	+2.79	+3.14	+2.63	+2.67	+3.38	+0.25	+6.68	–	–
XRPUSDT	obi	+3.53	189	+0.48	+0.48	+3.17	+2.45	+1.74	+3.09	+2.84	+2.67	+2.52	+3.63	+3.21	+1.70	+2.63	+2.76
BNBUSDC	glft	+1.72	489	+1.53	+1.23	+0.75	−1.17	−3.05	+0.58	+1.71	+1.53	+0.64	+2.50	+0.96	+1.05	+1.56	+1.01
BNBUSDC	hfgrid	+0.36	27	+0.04	−0.37	+0.27	+0.08	−0.11	+0.20	−0.14	+0.41	+0.28	+0.59	+0.49	+0.47	−0.45	−0.72

Held-out cumulative return, %, seven sessions. AVAXUSDC's OBI choice has κ = 0, so it has no κ neighbours.

- **SUIUSDC OBI is the only configuration in Part C that survives every stress except one**: it is positive without the crash, without its best session, without the rebate, at 4× latency and under both pessimistic queue models. Halving κ from 4 to 2 turns it to −0.90%, so the markout widening is doing real work.
- **AVAXUSDC OBI survives the costs but is one session**: without its best day it is −0.37%.
- **XRPUSDT OBI** behaves like XRPUSDC OBI: robust to costs, made on the crash.
- **BNBUSDC GLFT depends on the rebate** (−1.17% at a 1 bp fee).

8. Walk-forward selection

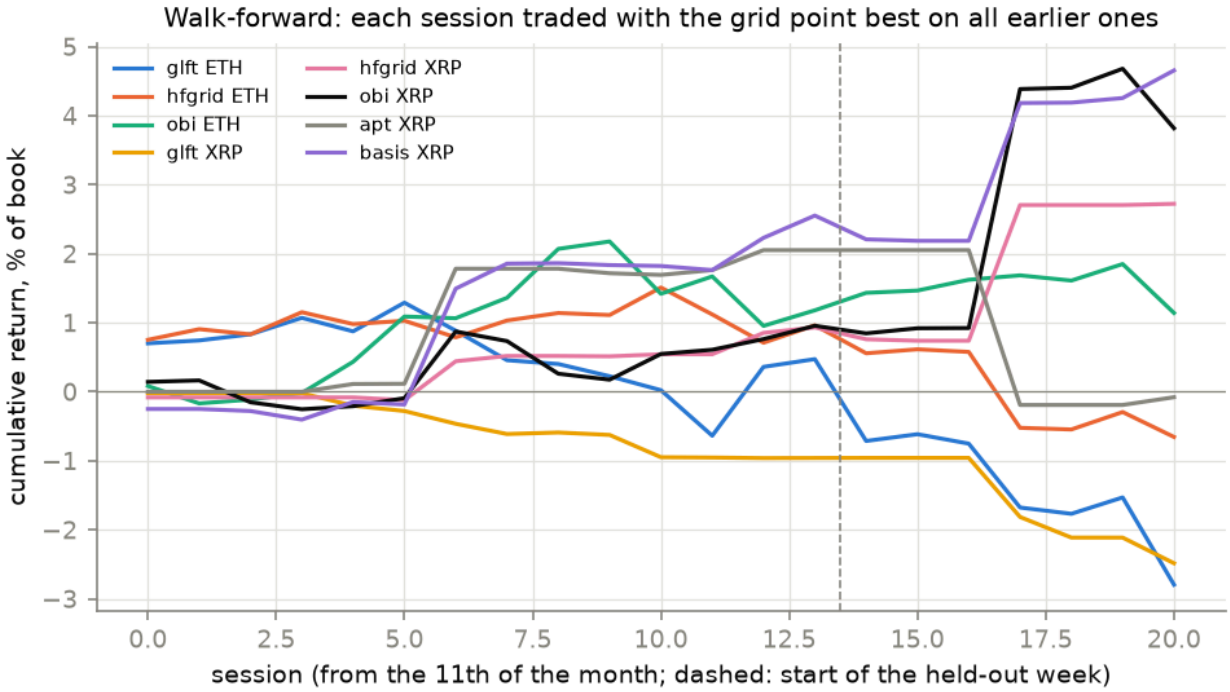


Figure — each session from the 11th of the month on, traded with the grid point that had the best return over every earlier session. Dashed: the start of the held-out week.

pair	walk-forward, 21 sessions	of which the held-out week	sessions up
XRP obi	+3.82	+2.83	14/21
XRP basis	+4.65	+2.05	10/21
XRP hfgrid	+2.72	+1.77	7/21
ETH obi	+1.14	−0.04	15/21
XRP apt	−0.08	−2.09	6/21
ETH hfgrid	−0.65	−1.60	10/21
XRP glft	−2.48	−1.54	4/21
ETH glft	−2.79	−3.25	9/21

Cumulative return, %. The walk-forward re-chooses every session, so it uses all the data a live system would have had, and no fixed split.

OBI XRP stays positive when chosen session by session over three weeks, not only on the one split, and ETH OBI is up on 15 of 21 sessions — though, per §6, largely on the rebate.

9. RL on top of the markout rule

The last follow-up: TD3 with C6's v2 recipe (residual policy, counterfactual reward, 5 s decisions), whose baseline is now the C8 grid choice — the markout rule with its chosen floor, κ and skew — and which also sees the rule's running toxicity (recipe `v2-mk`). Trained on the 24 training sessions, five seeds, on Kaggle CPU sessions.

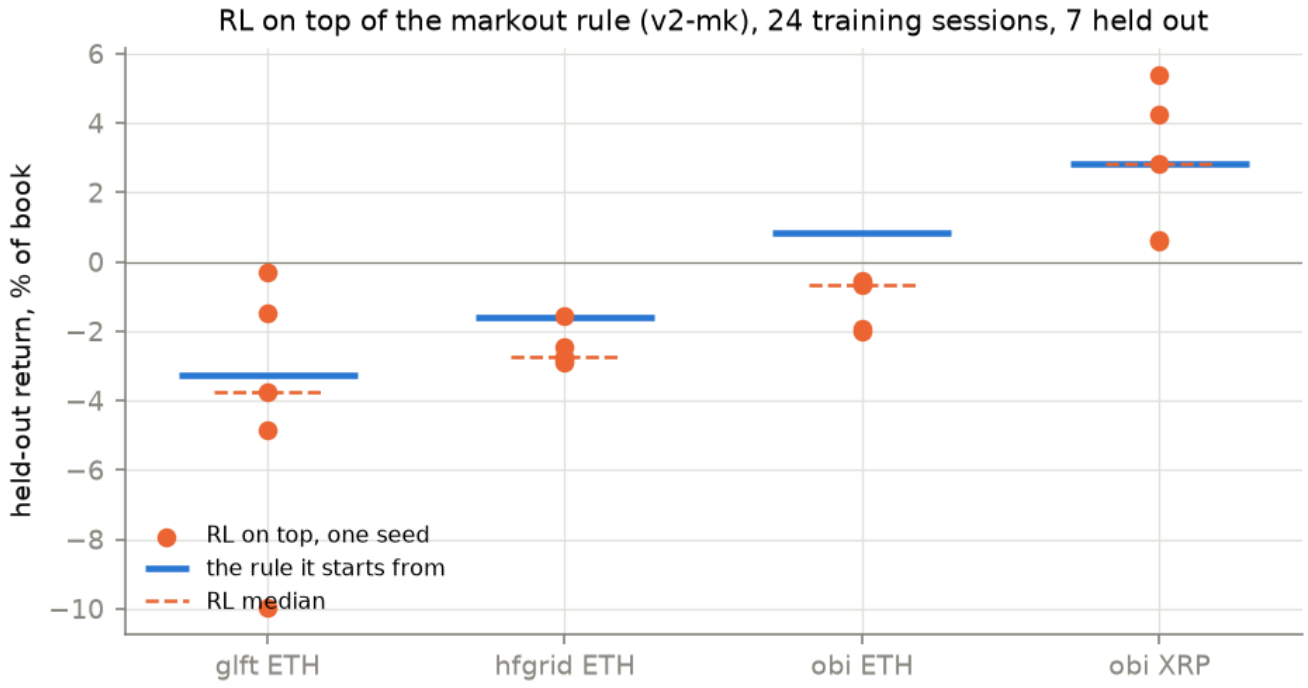


Figure — held-out return of each seed (dots) against the rule it starts from (solid) and the seeds' median (dashed).

asset	strategy	the rule (C8 grid)	RL on top, median of 5	seeds	rule ex-09-22	RL ex-09-22 (median)	RL fills/day
ETH	glft	−3.25	**−3.76**	−9.96, −1.48, −3.76, −4.85, −0.31	−2.34	−1.37	20–52
ETH	hfgrid	−1.60	**−2.75**	−1.57, −2.82, −2.75, −2.88, −2.44	−0.51	−0.74	23–74
ETH	obi	+0.85	**−0.67**	−1.94, −0.58, −2.00, −0.67, −0.54	+0.14	−0.82	22–1577
XRP	obi	+2.84	**+2.83**	+0.63, +4.24, +0.57, +5.40, +2.83	+0.09	+0.24	104–779

The policy never beats the rule it adjusts. On ETH it is worse on all three strategies; on OBI XRP the median ties and the seeds spread from +0.6 to +5.4 while trading up to seven times as often. This is the same answer C3, C6 and C4 gave for every other baseline, now on the strongest one.

10. So what makes money?

configuration	held out	ex 09-22	ex best day	fee 0	fee +2 bp	latency ×4	fills/day	verdict
SUIUSDC OBI (floor 8 bp, κ 4)	**+2.52**	**+1.77**	**+1.52**	**+2.05**	+0.21	**+1.30**	175	makes money on ordinary sessions, robust to costs; sensitive to κ
XRPUSDC OBI (floor 12 bp, κ 2, skew ×2)	+2.84	+0.09	+0.09	+2.61	+1.68	+1.17	106	robust to costs; the gain is the crash
XRPUSDT OBI (floor 8 bp, κ 2)	+3.53	+0.48	+0.48	+3.17	+1.74	+2.84	189	as XRPUSDC
AVAXUSDC OBI (floor 16 bp, κ 0)	+2.68	+2.34	−0.37	+2.35	+1.04	+3.14	150	one session
ETHUSDC OBI (floor 6 bp, κ 4)	+0.85	+0.14	+0.14	+0.09	−2.92	−0.48	369	the rebate
BNBUSDC GLFT (floor 2 bp, κ 1)	+1.72	+1.53	+1.23	+0.75	−3.05	+1.71	489	the rebate at higher fees

Held-out cumulative return, %, seven sessions (09-19 to 09-25).

The pattern across Part C holds and sharpens: **OBI with a wide, per-asset floor and the markout rule is the one strategy that makes money held out while trading**, on five of nine assets; **one of them, SUIUSDC, does so without the crash, without its best session and without the rebate**. Everything that tried to improve on it — more knobs (§5), a breaker (§4), RL (§9) — made it worse held out.

11. What would push further

- **The forward test** (the follow-up left out here): the frozen SUIUSDC and XRP OBI choices on sessions after 2025-09-25, with no re-selection. Seven held-out sessions cannot tell a +2.5% edge from luck, and three weeks of new data can.
- **OBI on more assets**, since it is the only strategy that generalises: the other SUI/ENA/AVAX quote currencies and any asset with comparable volume.
- **A crash-aware version of the breaker** — pause on volatility, but not when the position is already on the right side — since the breaker's cost on OBI XRP is exactly the crash session.
- **κ as a function of the asset's markout**, since SUIUSDC OBI's result depends on κ = 4 and no rule chooses κ yet except the grid.

12. How it was run, and what went wrong

- **Where.** About 36,000 session replays: ETHUSDC on this machine (four cores, 16 GB) and on Kaggle CPU sessions; XRPUSDC and the breadth tier on Kaggle; RL on Kaggle; the residual-LN network on a Colab T4. Every host writes the same (configuration, session) cache, merged by hash.
- **Memory.** Three ETH workers loading the same heavy session (09-24 and 09-25) at once exceeded 16 GB three times; the lost sessions were rerun on Kaggle (30 GB), which walks the exported list from the end.
- **A latency race.** Two Kaggle workers generated the same day's synthetic latency file at once and one read the other's half-written file; `config.ensure_latency` now writes to a temporary name and renames.
- **The Kaggle bridge.** Account calls failed with "Unauthenticated" behind a valid token because `mcp-remote` replayed Kaggle's anonymous session cookie (fixed in ds-agent, `--disable-cookies`).

13. Sources

what	source
the strategies, floors and presets	C6
the markout rule and the σ correction	C7
the grid, and why selection needs 20+ fills	C4
the RL recipe (v2)	C6 §8
queue models, latency, fees	hftbacktest 2.4.4 documentation; Binance USDⓈ-M fee schedule
code	`experiments/c8_risk_controls/` (`c8.py`, `report.py`), `experiments/c6_strategy_zoo/zoo.py`